Amin Tabrizian

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RESEARCH INTERESTS

Reliable learning-based decision making in safety-critical systems: large language models and agents, reinforcement learning under partial observability and delayed information, and scalable evaluation and red-teaming of AI systems.

EDUCATION

The George Washington University — **Washington, D.C.**
PhD in Computer Science. Advisor: Prof. Peng Wei — *Aug 2022 - May 2027 (Expected)*

University of Maryland, College Park — **College Park, MD**
Visiting Student in Computer Science — *2023 - 2024*

University of Tehran — **Tehran, Iran**
BSc in Mechanical Engineering — *2017 - 2021*

RESEARCH EXPERIENCE

Intelligent Aerospace Systems Laboratory, George Washington University — **Washington, D.C.**
Graduate Research Assistant — *Aug 2022 - Present*

- **LLM flight planning** (NASA-sponsored; with MIT Lincoln Laboratory and Honeywell): built the first end-to-end LLM flight planner for eVTOL/UAS under wind hazards, with 98% valid plans and about 2 km average deviation from A*. Extended it to an agentic system with RAG memory and a multimodal coach agent that critiques each plan
- **Delay-aware reinforcement learning** (with FHWA Saxton Laboratory): formulated highway on-ramp merging as a random-delay MDP and trained a reinforcement learning agent with a GRU delay-aware encoder and a physics-based safety controller; over 99% merge success with communication delays up to 2.0 s
- **Vision-language models**: co-developed text-encoder training on synthetic paragraph captions that matches Long-CLIP-L on ShareGPT4V and beats it by more than 14 points on DOCCI
- **LLM safety evaluation**: built red-teaming and stress-testing pipelines for jailbreaks, prompt injection and reasoning failures across several LLM families

WORK EXPERIENCE

Cignus Consulting — **Leesburg, VA**
AI Engineer Intern — *Jun 2026 - Present*

- Built a tool-calling LLM agent on AWS Bedrock with RAG and 20 tools, deployed with layered guardrails and monitoring
- Evaluated the agent on 695 test cases (94.1% accuracy) and a 200-case adversarial safety suite (200/200 passed)

PUBLICATIONS

- **A. Tabrizian**, Z. Huang, A. Aziz, P. Wei. Delay-Aware Reinforcement Learning for Highway On-Ramp Merging under Stochastic Communication Latency. *IEEE Internet of Things Journal*, 2026
- S. Chen, **A. Tabrizian**, P. Wei, Y. Wang, X. Qu. Adaptive Air Traffic Management via Large Language Model Augmented Reinforcement Learning. *Transportation Research Part E*, vol. 214, 2026
- M. Ghazanfari, **A. Tabrizian**, A. Aziz, B. Wang, P. Wei. A Paragraph is Worth a Thousand Captions: Rethinking Text Supervision for Vision-Language Retrieval. *arXiv preprint*, 2026
- M. Ghazanfari, **A. Tabrizian**, A. Mehrabian, P. Wei. EO-Agents: A Three-Agent LLM Pipeline for Earth Observation Hypothesis Generation. *ICML 2026 Workshop on AI for Science*
- T. Darrell, M. Ghazanfari, J. K. Kam, A. Bayen, **A. Tabrizian**, P. Wei. Towards Automated Air Traffic Safety Assessment Around Non-Towered Airports Using Large Language Models. *AIAA Aviation Forum*, 2026

- **A. Tabrizian**, M. Ghazanfari, P. Wei. Chain-of-Thought Flight Planner: End-to-End LLM Routing Under Wind Hazards. *AIAA Aviation Forum*, 2025. **Best Student Paper Award**, Air Transportation Systems
- **A. Tabrizian**, P. Gupta, A. Taye, J. Jones, E. Thompson, S. Chen, T. Bonin, D. Eberle, P. Wei. Using Large Language Models to Automate Flight Planning under Wind Hazards. *IEEE/AIAA DASC*, 2024
- P. Razzaghi, **A. Tabrizian**, W. Guo, S. Chen, A. Taye, E. Thompson, A. Bregeon, A. Bahari, P. Wei. A Survey on Reinforcement Learning in Aviation Applications. *Engineering Applications of Artificial Intelligence*, vol. 136, 2024
- C. Vu, M. Ghazanfari, K. Dong, A. Taye, **A. Tabrizian**, P. Wei. Transformer or CNN? Benchmarking RT-DETR and YOLOv8 for Small UAS Autonomous Landing. *AIAA Aviation*, 2025
- J. Xu, C. Vu, A. Taye, **A. Tabrizian**, P. Wei. Small UAS Landing Site Detection with ArUco Markers and Deep Learning Based Computer Vision. *AIAA Aviation*, 2025
- D. Ding, C. Vu, A. Taye, **A. Tabrizian**, P. Wei, Z. Zhao. Synthetic Data Generation for Computer Vision Based Autonomous Landing for Small UAS Package Delivery. *AIAA Aviation*, 2025
- A. Famili, **A. Tabrizian**, T. Atalay, A. Stavrou. RADIO: Reinforcement Learning-Aided Deployment of Wi-Fi Routers in 5G Networks for Indoor Drone Orchestrating. *IEEE iThings*, 2024
- A. Famili, **A. Tabrizian**, T. Atalay, A. Stavrou, P. Wei. RAPID: Reinforcement Learning-Aided Femtocell Placement for Indoor Drone Localization. *IEEE ICCCN*, 2024
- A. Famili, T. Atalay, **A. Tabrizian**, A. Stavrou. GREEN: Precise Geolocation in Metaverse using Reinforcement Learning-Enabled Sensor Placement. *IEEE MetaCom*, 2024

INVITED TALKS

- Honeywell Advanced Technology (Dec 2025); INFORMS Annual Meeting, Atlanta (2025); AIAA Aviation, Las Vegas (2025); AAAI 2025 Spring Symposium, San Francisco; INFORMS Annual Meeting, Seattle (2024); IEEE ICCCN, Big Island (2024)

TEACHING

Large Language and Vision Models (MAE 6291 / CSCI 6907) — **Washington, D.C.**
Instructor, The George Washington University — *Fall 2026*

Linear System Dynamics (MAE 3134) — **Washington, D.C.**
Teaching Assistant, The George Washington University — *Spring 2026*

University of Tehran — **Tehran, Iran**
Teaching Assistant (5 courses): Probability, Control, AI, Instrumentation, Interactive Learning — *2020 - 2021*

SERVICE

- **Organizer**: NASA System-Wide Safety Program Workshop (2025); Safe and Reliable AI Workshop (2024)
- **Reviewer**: Journal of Aerospace Information Systems (2022-2025); AIAA Aviation (2025); Engineering Applications of AI (2024)

TECHNICAL SKILLS

- **Languages & Tools**: Python, C/C++, SQL, MATLAB/Simulink, Git, Linux, Docker, AWS Bedrock
- **Deep Learning**: PyTorch, TensorFlow, Hugging Face; transformers, CNNs, GRU encoders, CLIP-style contrastive training, RT-DETR/YOLOv8
- **LLMs & Agents**: tool-calling and multi-agent systems, RAG, chain-of-thought, guardrails, benchmark evaluation, red-teaming
- **Reinforcement Learning**: actor-critic methods, delayed and partially observable MDPs, safe RL